

TE Curve Verification Pipeline Best Model Collage Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative `TE Curve Verification Pipeline` TE-curve predictions for the current best reference, RCIM Model-Bank Reproduction, Wave 1 directional, and Wave 1 global models. Each model is shown as one four-image collage so local oscillation tracking can be inspected directly.

Scope

- each collage contains four deterministic held-out test curves;
- forward models are shown on forward curves only;
- backward models are shown on backward curves only;
- global Wave 1 models are shown on two forward and two backward curves;
- `Measured TE` uses the same line width as predictions and a dark-gray color for balanced visual comparison.

Metrics Summary

Forward Accepted Non PINN Incumbent Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
<code>accepted_periodic_gru_sequence_fw</code>	<code>accepted_non_pinn_incumbent</code>	Fw	0.001618	0.001931	3.278
<code>accepted_periodic_mlp_harmonic_fw</code>	<code>accepted_non_pinn_incumbent</code>	Fw	0.001694	0.002008	3.439

Backward Accepted Non PINN Incumbent Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
<code>accepted_periodic_gru_sequence_bw</code>	<code>accepted_non_pinn_incumbent</code>	Bw	0.001837	0.002196	3.494
<code>accepted_periodic_mlp_harmonic_bw</code>	<code>accepted_non_pinn_incumbent</code>	Bw	0.001912	0.002263	3.581

Global Accepted Non PINN Incumbent Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
accepted_periodic_gru_sequence_global	accepted_non_pinn_incumbent	global	0.001810	0.002141	3.567
accepted_periodic_mlp_harmonic_global	accepted_non_pinn_incumbent	global	0.001734	0.002054	3.385

Global Wave52r Integrated Specialist Trained Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_integrated_a02_seed_314159	wave52r_integrated_specialist_trained	global	0.001462	0.001735	2.890
wave52r_integrated_a03_seed_161803	wave52r_integrated_specialist_trained	global	0.001457	0.001731	2.885
wave52r_integrated_a04_seed_161803	wave52r_integrated_specialist_trained	global	0.001457	0.001729	2.877
wave52r_integrated_a05_seed_161803	wave52r_integrated_specialist_trained	global	0.001461	0.001733	2.884
wave52r_integrated_a06_seed_271828	wave52r_integrated_specialist_trained	global	0.001463	0.001736	2.889
wave52r_integrated_a07_seed_314159	wave52r_integrated_specialist_trained	global	0.001463	0.001736	2.888
wave52r_integrated_a07_seed_161803	wave52r_integrated_specialist_trained	global	0.001463	0.001736	2.890

Forward Wave52r Offline Leader Cross Surface Promotion Models

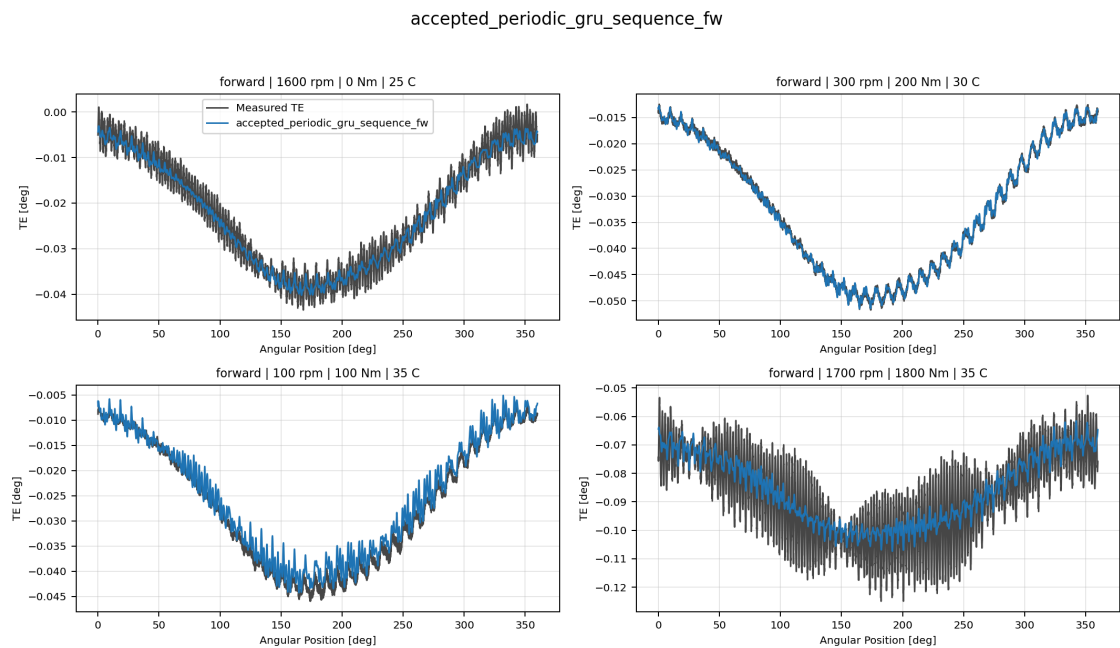
Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_promotion_h08_fw_seed_161803	wave52r_offline_leader_cross_surface_promotion	Fw	0.001688	0.001985	3.477

Global Wave52r Offline Leader Cross Surface Promotion Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_promotion_k01_global_seed_271828	wave52r_offline_leader_cross_surface_promotion	global	0.001464	0.001738	2.893

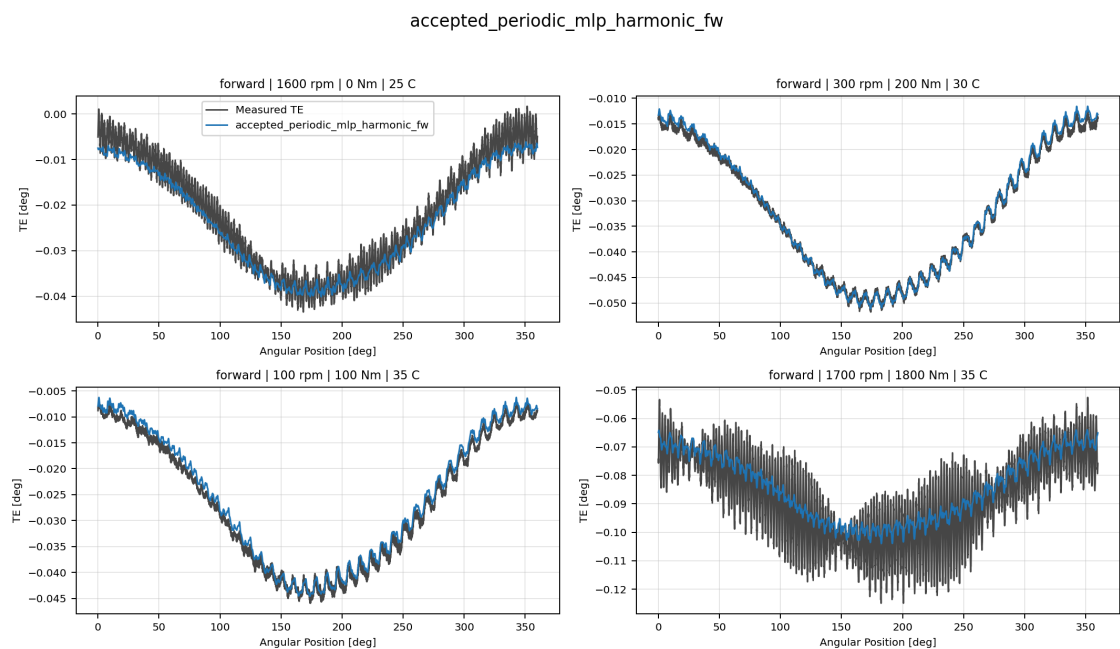
Collage Gallery - Forward Accepted Non PINN Incumbent Models

accepted_periodic_gru_sequence_fw



accepted_periodic_gru_sequence_fw TE Curve Verification Pipeline collage

accepted_periodic_mlp_harmonic_fw

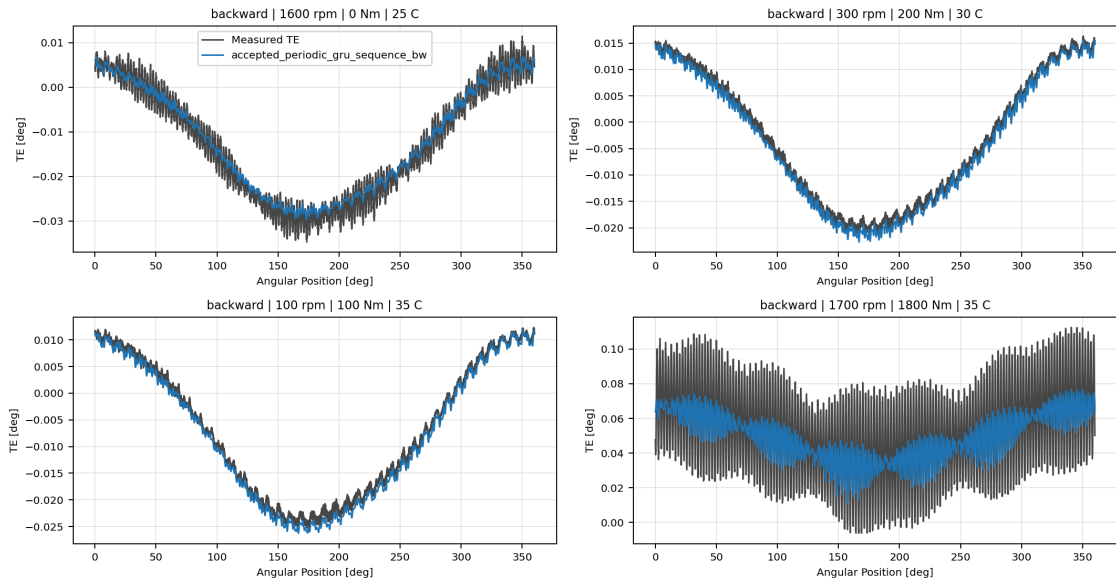


accepted_periodic_mlp_harmonic_fw TE Curve Verification Pipeline collage

Collage Gallery - Backward Accepted Non PINN Incumbent Models

accepted_periodic_gru_sequence_bw

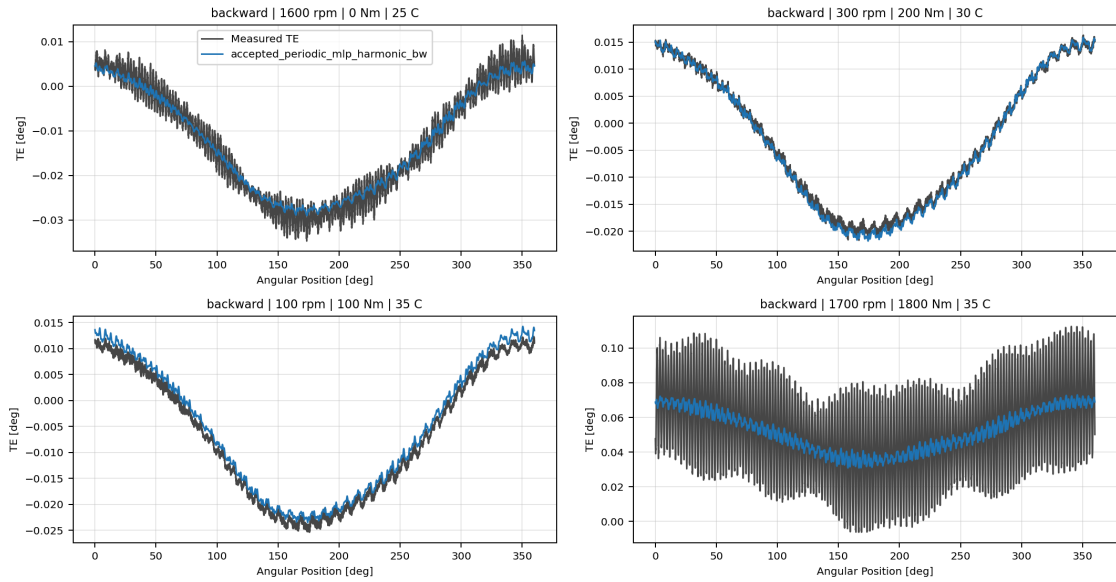
accepted_periodic_gru_sequence_bw



accepted_periodic_gru_sequence_bw TE Curve Verification Pipeline collage

accepted_periodic_mlp_harmonic_bw

accepted_periodic_mlp_harmonic_bw

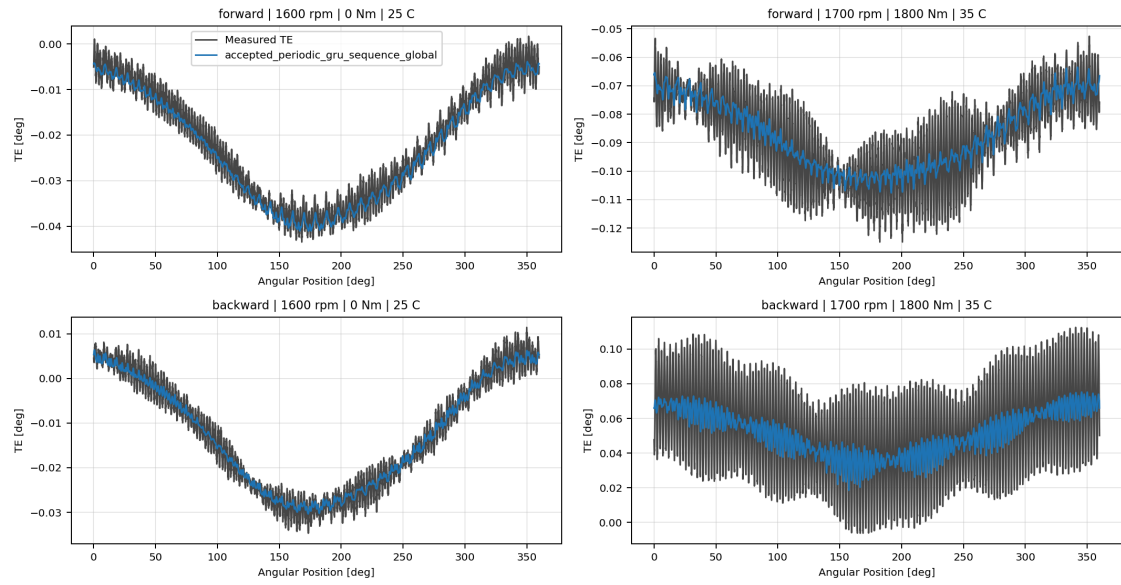


accepted_periodic_mlp_harmonic_bw TE Curve Verification Pipeline collage

Collage Gallery - Global Accepted Non PINN Incumbent Models

[accepted_periodic_gru_sequence_global](#)

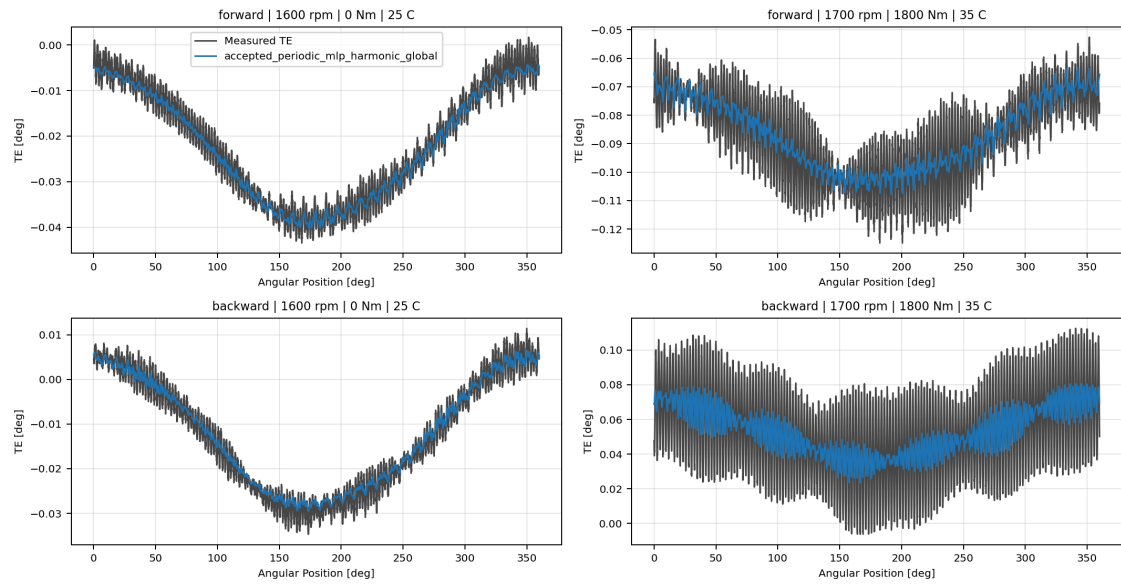
accepted_periodic_gru_sequence_global



[accepted_periodic_gru_sequence_global TE Curve Verification Pipeline collage](#)

[accepted_periodic_mlp_harmonic_global](#)

accepted_periodic_mlp_harmonic_global

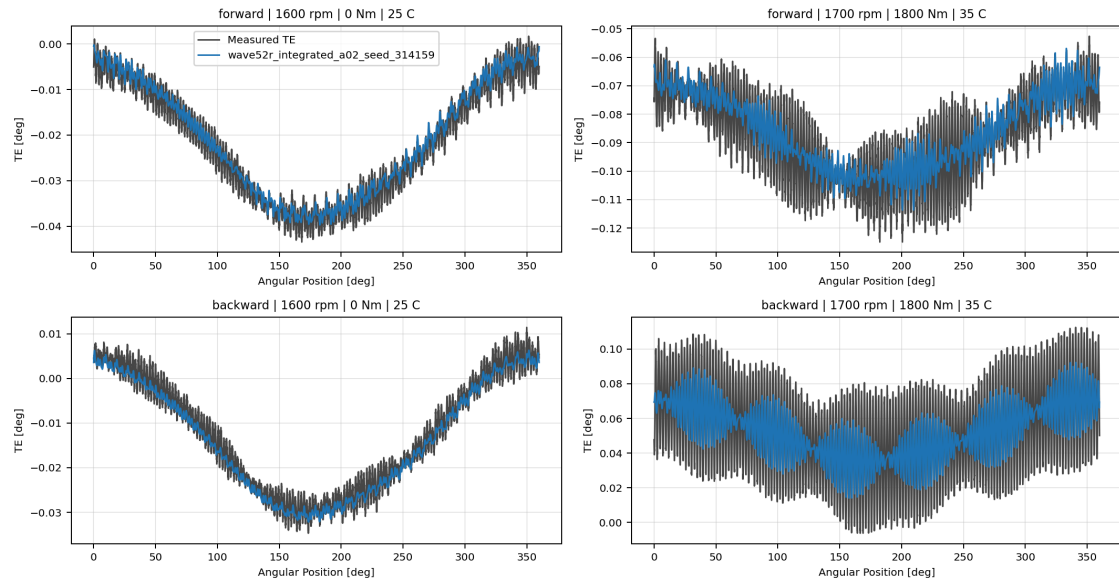


[accepted_periodic_mlp_harmonic_global TE Curve Verification Pipeline collage](#)

Collage Gallery - Global Wave52r Integrated Specialist Trained Models

[wave52r_integrated_a02_seed_314159](#)

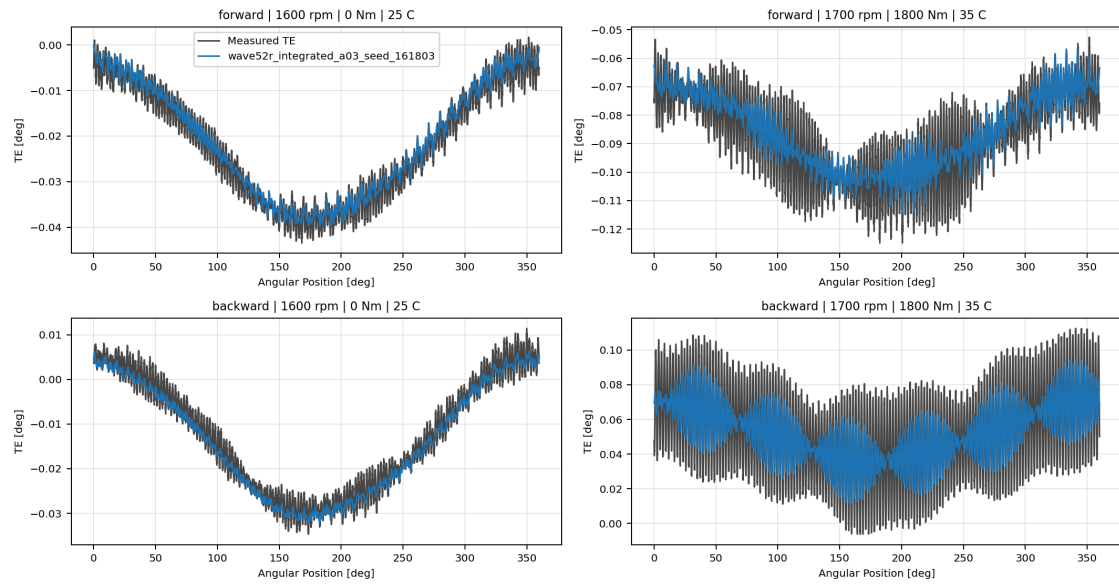
wave52r_integrated_a02_seed_314159



wave52r_integrated_a02_seed_314159 TE Curve Verification Pipeline collage

[wave52r_integrated_a03_seed_161803](#)

wave52r_integrated_a03_seed_161803



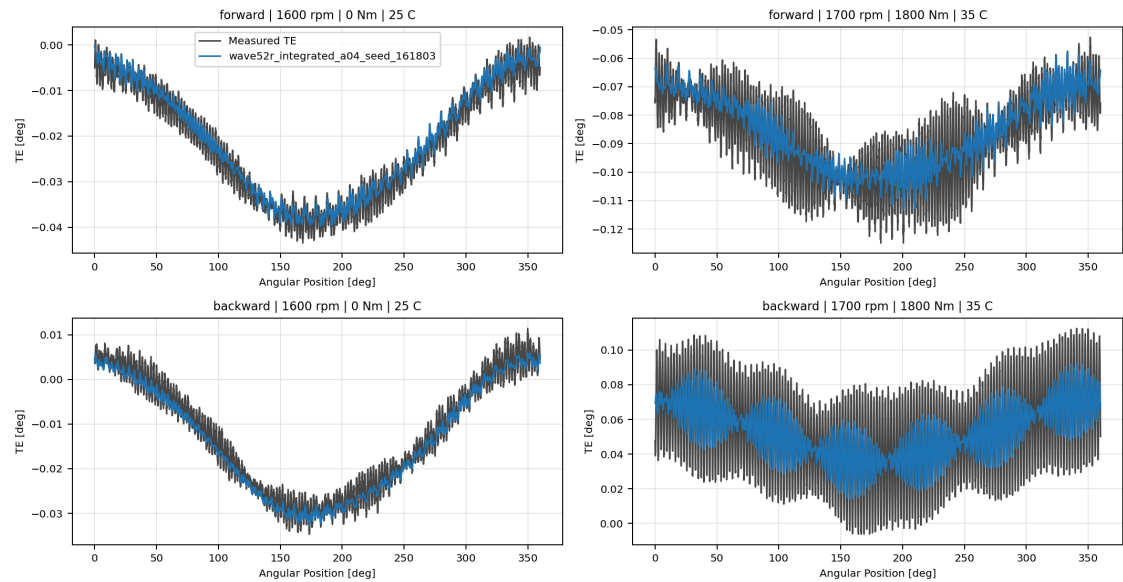
wave52r_integrated_a03_seed_161803 TE Curve Verification Pipeline collage

Collage Gallery - Global Wave52r Integrated Specialist Trained Models

Continued

wave52r_integrated_a04_seed_161803

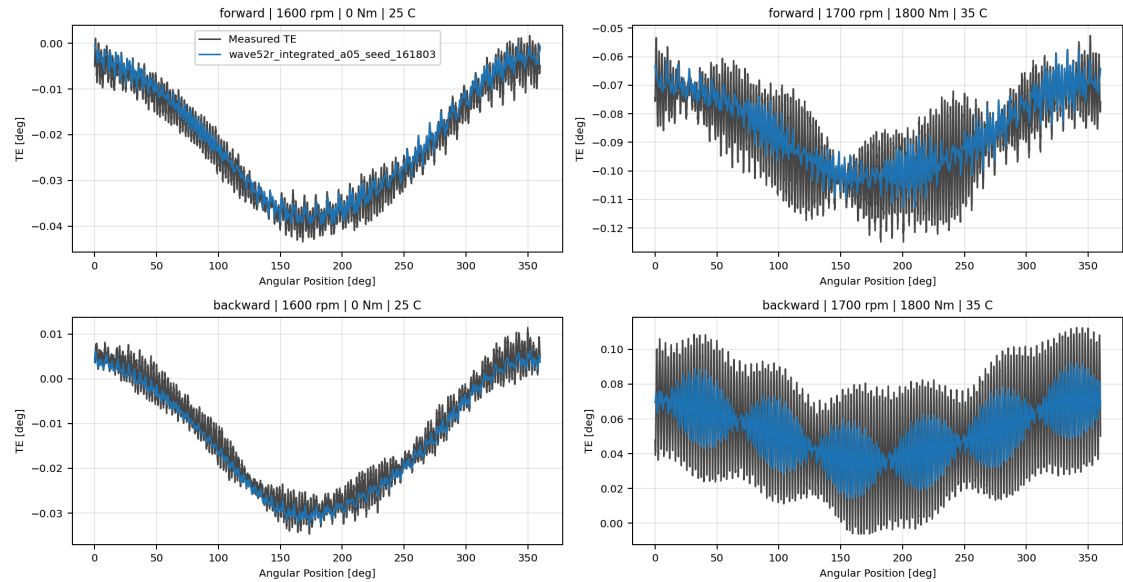
wave52r_integrated_a04_seed_161803



wave52r_integrated_a04_seed_161803 TE Curve Verification Pipeline collage

wave52r_integrated_a05_seed_161803

wave52r_integrated_a05_seed_161803

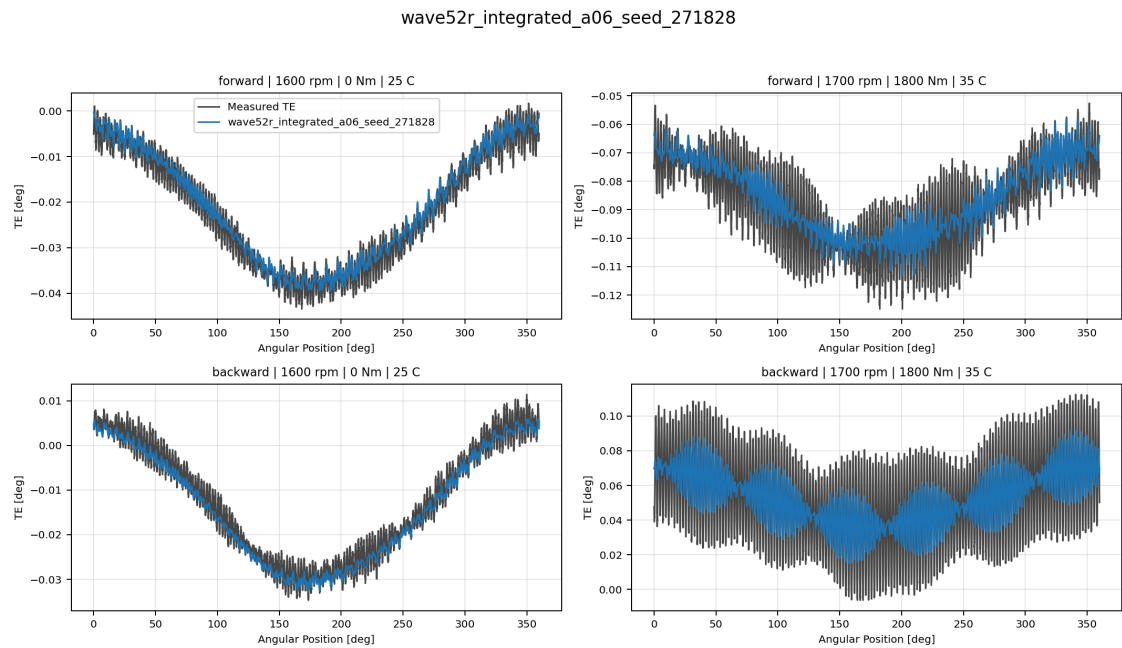


wave52r_integrated_a05_seed_161803 TE Curve Verification Pipeline collage

Collage Gallery - Global Wave52r Integrated Specialist Trained Models

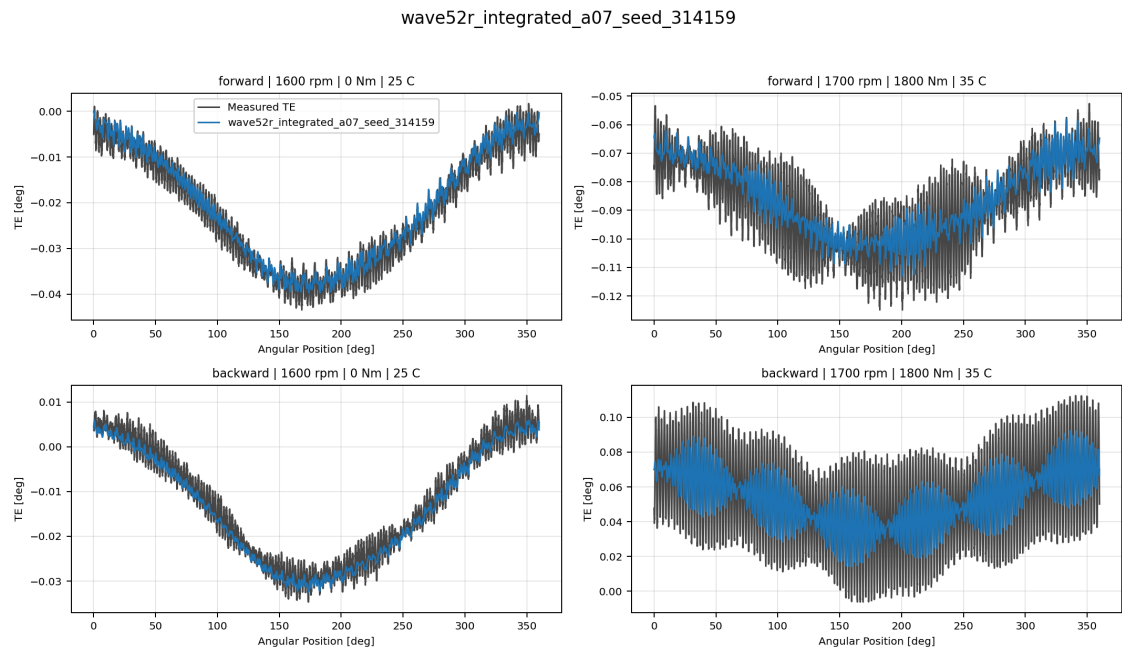
Continued 2

wave52r_integrated_a06_seed_271828



wave52r_integrated_a06_seed_271828 TE Curve Verification Pipeline collage

wave52r_integrated_a07_seed_314159

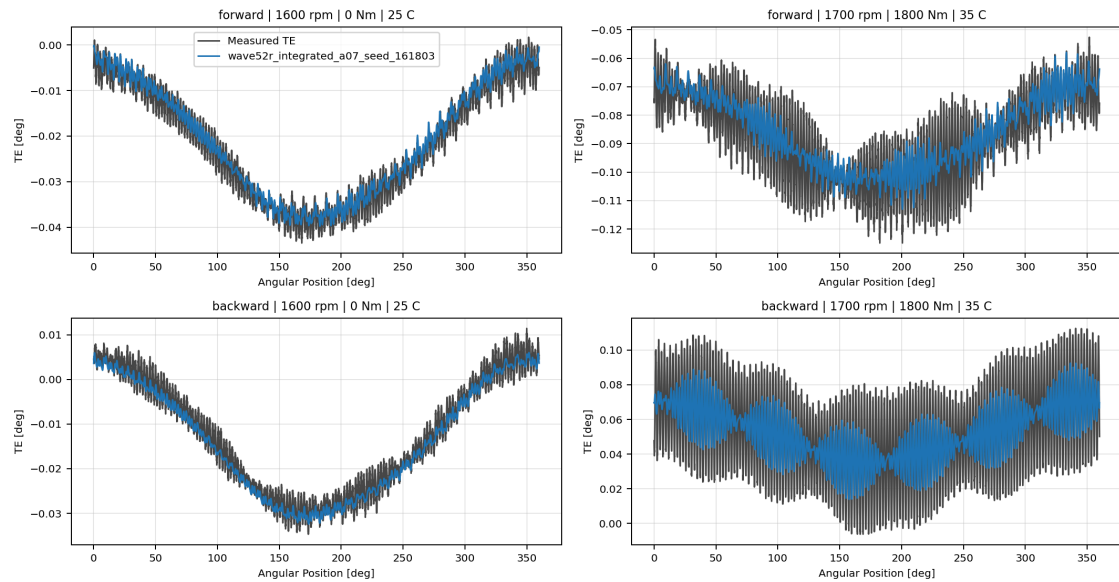


wave52r_integrated_a07_seed_314159 TE Curve Verification Pipeline collage

Collage Gallery - Global Wave52r Integrated Specialist Trained Models Continued 3

[wave52r_integrated_a07_seed_161803](#)

wave52r_integrated_a07_seed_161803

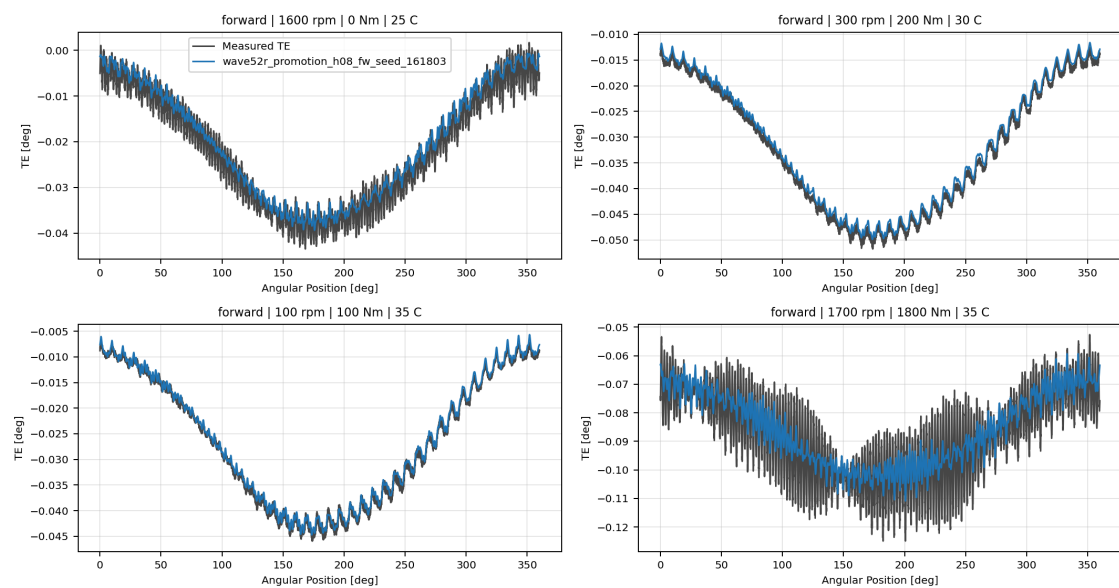


wave52r_integrated_a07_seed_161803 TE Curve Verification Pipeline collage

Collage Gallery - Forward Wave52r Offline Leader Cross Surface Promotion Models

[wave52r_promotion_h08_fw_seed_161803](#)

wave52r_promotion_h08_fw_seed_161803

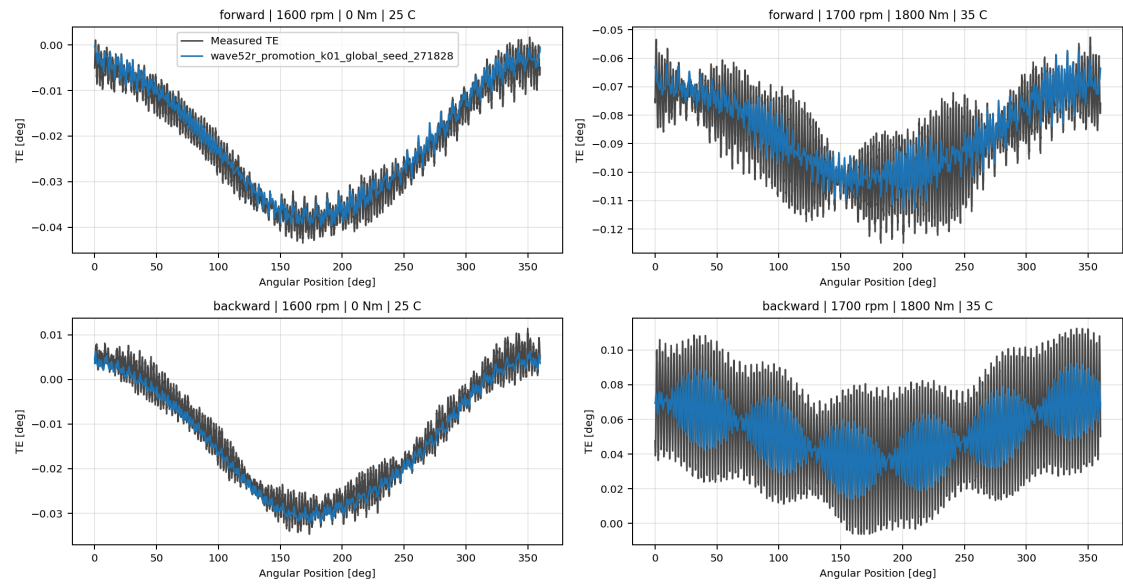


wave52r_promotion_h08_fw_seed_161803 TE Curve Verification Pipeline collage

Collage Gallery - Global Wave52r Offline Leader Cross Surface Promotion Models

wave52r_promotion_k01_global_seed_271828

wave52r_promotion_k01_global_seed_271828



wave52r_promotion_k01_global_seed_271828 TE Curve Verification Pipeline collage

Output Artifacts

- output directory:
`output\validation_checks\wave52r_integrated_specialist_track2_best_model_collage_report\2026-08-04-00-33-58__track2_best_model_collage_report ;`
- summary YAML:
`output\validation_checks\wave52r_integrated_specialist_track2_best_model_collage_report\2026-08-04-00-33-58__track2_best_model_collage_report\track2_best_model_collage_summary.yaml ;`
- metrics CSV:
`output\validation_checks\wave52r_integrated_specialist_track2_best_model_collage_report\2026-08-04-00-33-58__track2_best_model_collage_report\track2_best_model_collage_metrics.csv ;`
- report Markdown:
`doc\reports\analysis\te_curve_verification_pipeline\02_visual_reports\wave52r_integrated_specialist_best_model_collage_report\[2026-08-04]\track2_best_model_collage_report.md .`